# Conditional probability

## Lecture and worked examples

Maths Explorer

## Reference course

In a weighted tree, the probabilities leaving the same node sum to 1. We multiply the probabilities along a path: $P(A \cap B) = P(A)P_A(B)$. We add the probabilities of mutually exclusive paths leading to the event we are looking for.

## Reference course

Do not confuse $P_A(B)$ and $P_B(A)$. The condition comes after the word “given” and is written as a subscript in this notation.

## Reference course

40% of customers buy online. Among them, 12% return their order; among the others, 5% return it. Then $P(R) = 0.40 \times 0.12 + 0.60 \times 0.05 = 0.078$. The overall return rate is therefore 7.8%.

# Conditioning

- Define a conditional probability.
- Move from counts to probabilities.
- Interpret.

## Conditioning

In a survey, 180 customers out of 600 use a premium card and 72 of these 180 customers buy every week. Calculate $P(P)$, $P(P \cap H)$ and $P_P(H)$, then interpret.

## Conditioning

For $P(A) > 0$, $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. This is the probability of $B$ within the population restricted to $A$.

## Conditioning

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

# Conditioning

**One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

**Two-hour plan:** add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

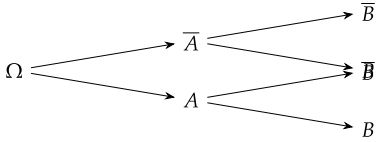
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

# Weighted trees

- Construct and complete a tree.
- Read a conditional probability.
- Check the sums.

# Weighted trees

35% of orders are express. Among them, 4% are late; among standard orders, 9% are late. Construct and complete the tree.



## Weighted trees

In a tree, the branches leaving a node sum to 1. The second-level probabilities are conditional.

## Weighted trees

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## Weighted trees

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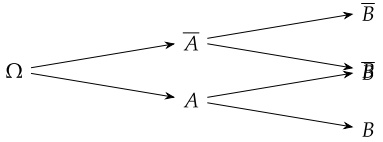
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## Product of probabilities along a path

- Calculate an intersection.
- Combine paths.
- Translate an event.

# Product of probabilities along a path

Using the tree from session 40, calculate the probability of an order being express and late, then the total probability of a delay. Interpret both results.



## Product of probabilities along a path

We multiply along a path: $P(A \cap B) = P(A)P_A(B)$. We add the mutually exclusive paths that realise the event.

## Product of probabilities along a path

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

## Product of probabilities along a path

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# Independence

- Test independence.
- Compare probabilities.
- Justify a conclusion.

## Independence

In a population, $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.12$. Test independence in two ways. Then change $P(A \cap B)$ to 0.15 and start again.

## Independence

$A$ and $B$ are independent if $P(A \cap B) = P(A)P(B)$, equivalently if $P_A(B) = P(B)$ when $P(A) > 0$.

## Independence

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## Independence

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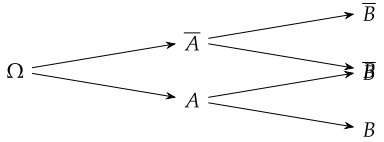
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# Partition and total probability

- Recognise a partition.
- Apply total probability.
- Organise the calculation.

# Partition and total probability

A factory has three lines A, B, C producing 20%, 50% and 30% of the items respectively. Their defect rates are 1%, 2% and 4%. Calculate the overall defect rate.



## Partition and total probability

If $(A_i)$ is a partition, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. Each term corresponds to a path leading to $B$.

## Partition and total probability

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

# Partition and total probability

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## Problems in context

- Choose a tree or a table.
- Solve a problem in context.
- Interpret.

## Problems in context

Turn a table of counts (customers/subscription/cancellation) into a tree, then answer a question in each direction: probability of cancelling given the subscription; probability of having the subscription given the cancellation.

## Problems in context

Choosing between a tree and a table: the tree highlights the order of conditioning; the table suits counts and reversed readings well.

## Problems in context

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## Problems in context

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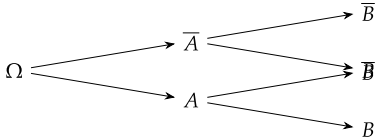
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# Mixed practice

- Calculate a reverse probability.
- Chain total probability and conditioning.
- Check the result.

# Mixed practice

In the situation of session 43, given that an item is defective, calculate the probability that it comes from line C. Explain why this result is not 30%.



## Mixed practice

For a reverse probability, first calculate the total probability of the conditioning event, then use the definition of conditional probability.

## Mixed practice

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## Mixed practice

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## Review and assessment

- Use all the methods.
- Write up with the notation.
- Remedy mistakes.

## Review and assessment

Complete a four-station circuit: complete a tree; calculate an intersection; apply the total probability formula; test independence. Each group leaves a written justification for the next group.

## Review and assessment

Always name the events, complete all the branches, write the formula before calculating and interpret the final probability.

## Review and assessment

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